

# Machine Design Assignment

## Analysis of Helicopter Rotor Shaft

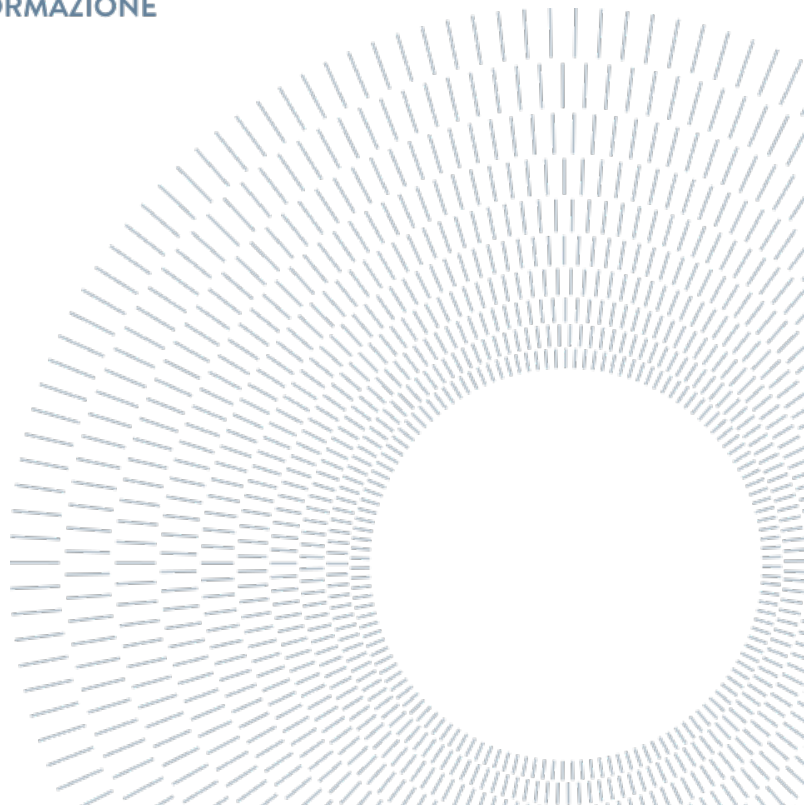
| Name   | Surname   | Student ID |
|--------|-----------|------------|
| Evan   | Emberger  | ██████     |
| Ernest | Yakubayev | ██████     |
| Oscar  | Pratola   | ██████     |
| Talha  | Nadeem    | ██████     |

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# 1 Introduction

## 1.1 Problem Statement

Rotor shafts for helicopters are essential parts that are essential to the design and functionality of helicopters. They act as the hub around which the rotor blades revolve, providing lift and allowing the helicopter to navigate. The rotor assembly receives power from the engine via the rotor shafts, which enables the blades to spin rapidly and generate the required lift for flight.

Rotor shafts are crucial because of their capacity to tolerate difficult operating conditions while effectively transferring power. As the rotor blades revolve, significant centrifugal forces, bending moments, and torsional stresses are placed on the rotor shafts. High-strength materials must be used in their design and construction to maintain structural integrity and prevent failure while in flight. Thorough inspections and maintenance procedures are required to keep the rotor shafts reliable and safe. Regular inspections assist in identifying any indications of wear, fatigue, or deterioration that can jeopardize the shafts' structural integrity [1].

As such, it can be said that a design assessment of the rotor shaft must be conducted. The design assessment includes the static assessment as well as the fatigue assessment of the mast in hand to verify its safety and structural integrity. At the same time, verification of the analytical assessment is to be checked with experimental results as well as comparisons to the numerical assessment.

## 1.2 Data Acquired

The relevant data needed for the assessment for structural analysis of the mast and the specimen included the geometry, loading conditions, supports, and material properties which were provided in the detail of the guidelines. The data application is further utilized for static and fatigue assessment and FEM assessment. In regards to the experimental data, the structural testing for the given specimen was conducted at CLASD laboratories. The data collected through the DAQ system provided an output file of relevant strains through the strain gauge rosette.

## 1.3 Workstation

The numerical investigation, in particular, required an operatable and optimum workstation to make the computations possible. A single workstation was utilized in the working for consistency of results. The following is a description of the workstation used for FEM computation using ABAQUS.

|                  |                                                  |
|------------------|--------------------------------------------------|
| <b>Processor</b> | AMD Ryzen 9 5900HX with Radeon Graphics 3.30 GHz |
| <b>RAM</b>       | 16.0 GB                                          |
| <b>System</b>    | 64-bit operating system                          |
| <b>GPU</b>       | Nvidia Geforce RTX 3070                          |

Table 1: Workstation Details

## 2 Static Assessment of the Mast

In this section, a simplified representation of the helicopter main rotor shaft is statically assessed to verify its resistance under operational loading conditions. subsection 2.1 begins by setting-up the problem as well as identifying reaction forces and internal load distributions. The critical section will then be identified in subsection 2.2 and assessed in subsection 2.3 and subsection 2.4.

Figure 1 represents the free body diagram of the mast to be analysed with the external forces in red and the reaction forces in blue. The shear force  $T$  and the bending moment  $M_f$  that can be seen in the figure have been placed such as to create the most critical loading condition for the mast, i.e: the condition leading to the highest stress. The reasoning for such placement is as follows:

- The cross-section of the mast is perfectly circular which means that only the relative placement of the loads with respect to each others matter for the state of stress of the mast.
- Considering that the beam is slender, the shear stress generated by the shear force is likely to be small in comparison to the bending stress induced by the shear force. The focus is thus put on the bending stress.
- For the bending stress to be maximum, the bending moment must be maximized which implies that the shear force must produce a moment of the same sign than the point moment  $M_f$  along any point of the mast. This is obtained when  $T$  and  $M_f$  are placed such as in Figure 1. Note that since the stress induced by the axial force and the torque are uniform along a radial position of the cross-section, the shear force and the bending moment could have been placed in the opposite direction than shown in the figure, still leading to the same result.
- Note: although it is specified that the loads are applied in the  $yz$  plane, it would be interesting to consider a loading case where  $T$  is out of that plane, allowing to add an extra torsion component to increase the criticality of the condition.

The mast geometry and external loading are defined as follows:

| Variable | Value   | Unit |
|----------|---------|------|
| $N$      | 100 000 | N    |
| $M_t$    | -80 000 | Nm   |
| $T$      | 32 000  | N    |
| $M_f$    | 35 000  | Nm   |
| $L_1$    | 0.438   | m    |
| $L_2$    | 0.202   | m    |

Table 2: Geometric and loading data of the static mast analysis

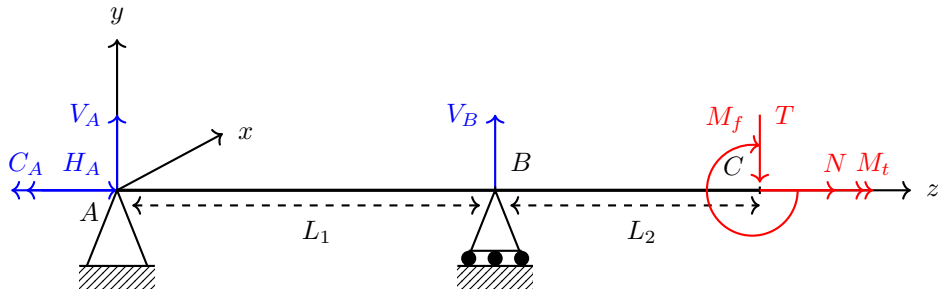


Figure 1: Free body diagram of the mast. Reaction and external loads are displayed in blue and red, respectively

## 2.1 Internal Load Distributions

Once the problem has been defined, the reaction forces and moments can be computed using Equations 1-4.

$$\Sigma F_z = 0 : H_A + N = 0 \rightarrow H_A = -N = 100000[N] \quad (1)$$

$$\Sigma M_z = 0 : M_t - C_A = 0 \rightarrow C_A = M_t = -80000[Nm] \quad (2)$$

$$\Sigma M_{x_B} = 0 : V_A \cdot L_1 + T \cdot L_2 + M_f = 0 \rightarrow V_A = \frac{-M_f - T \cdot L_2}{L_1} = -94666[N] \quad (3)$$

$$\Sigma F_y = 0 : V_A + V_B - T = 0 \rightarrow V_B = T - V_A \rightarrow V_B = 126666[N] \quad (4)$$

With the external and reaction loads defined, the internal loading diagrams of the mast are obtained and shown in Figure 2. The convention used for determining the sign of the internal loads are displayed in Figure 3.

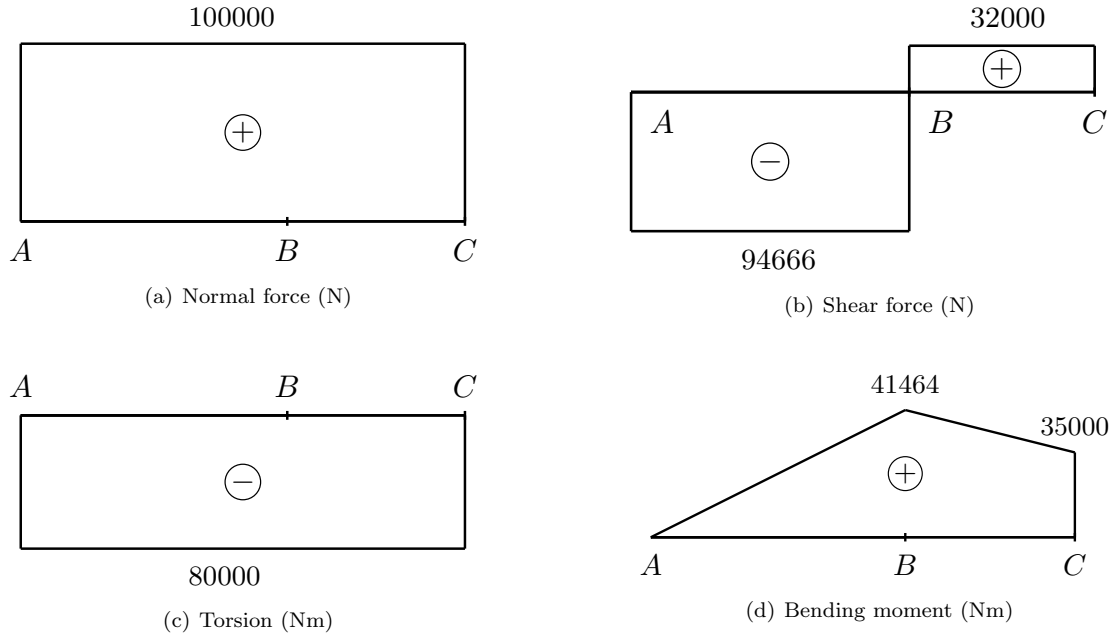


Figure 2: Internal loading diagram of the mast

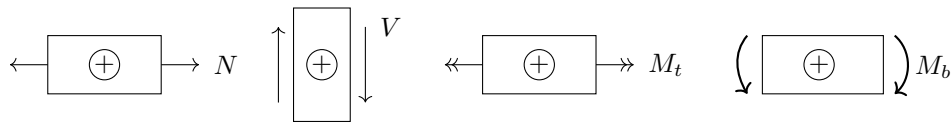


Figure 3: Loading convention used for the internal loading diagram of the mast

## 2.2 Critical Section

Internal normal force and torsion are constant along the mast's axis. Additionally, the shear force and the bending moment are both maximum at point B and remain very high in the region around it. Moreover, Figure 4 indicates that both sides around point B experience a reduction in shaft diameter. As such a geometry change is accompanied with a stress concentration factor and that a diameter reduction leads to a decrease of the cross-sectional area as well as of the moment of inertia and polar moment of inertia, the points slightly before and after point B will experience a greater stress than point B itself. As the internal shear force is significantly higher on the left side of point B than on its right side, the most critical section is located there as indicated in Figure 4.

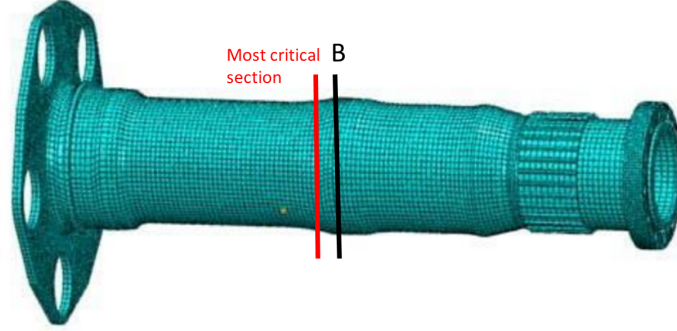


Figure 4: 3D visual representation of the mast. The critical section is indicated in red while the section referred as "B" in this analysis is highlighted in black

## 2.3 Principal and Equivalent Stresses

The analysis now focuses on the most critical section of mast, highlighted in red in Figure 4. No geometrical information about the diameter reduction is given. Therefore, it is difficult to estimate the exact position of the critical section and the loads acting on it. As a conservative approximation, the internal loads of point B will be used instead.

The relevant loading data are issued from the internal loading diagram at point B and, along with given geometrical data of the section, are presented in Table 3.

| Variable | Value   | Unit |
|----------|---------|------|
| $N$      | 100 000 | [N]  |
| $M_b$    | 41 464  | [Nm] |
| $M_t$    | -80 000 | [Nm] |
| $V$      | 94 666  | [N]  |
| $R_{in}$ | 0.07    | [m]  |
| $t$      | 0.009   | [m]  |

Table 3: Geometrical and loading data for the static analysis of the most critical section of the mast

The first step is determining the most loaded point of the section. Figure 5 shows a view of the cross-section of the most critical section of the mast along with the loads applied on it. The point of maximum tensile

bending stress occurs at point vertically farthest from the axis of application of the bending moment. As the normal force considered applies a uniform tensile stress on the cross-section, the point of tensile bending stress should be considered as a potential most loaded point. This corresponds to the "Point 1" highlighted in blue in Figure 5.

However point 1 does not show a stress contribution from the shear force and, as the shear load applied is significant, it may be interesting to consider it. This is done by considering "Point 2" in Figure 5, located where the shear force's shear stress contribution is the highest (although the bending stress will be zero there). Both points 1 and 2 are located on the external radius of the cross-section as the farthest points from the center are subjected to highest torsional shear stress, in addition to the bending stress already mentioned.

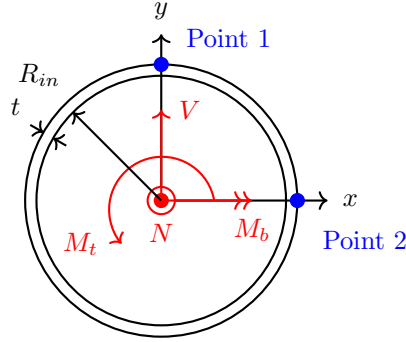


Figure 5: Diagram of the loaded critical cross-section of the mast (N and the z axis are out of plane)

The stresses are now computed to analyse points 1 and 2. Due to the normal force, for both points 1 and 2:

$$\sigma_{z,uniform} = \frac{N}{A} = \frac{100000}{\pi((R_{in} + t)^2 - R_{in}^2)} = 23.7MPa \quad (5)$$

Due to torsion, for both points 1 and 2:

$$\tau_{max} = \frac{M_t \cdot R_{max}}{J} = \frac{-80000 \cdot (R_{in} + t)}{\frac{\pi}{2}((R_{in} + t)^4 - R_{in}^4)} = 269.3MPa \quad (6)$$

Due to bending, at point 1:

$$\sigma_{z,bend} = \frac{M_b \cdot y}{I_{xx}} = \frac{41464 \cdot (R_{in} + t)}{\frac{\pi}{4}((R_{in} + t)^4 - R_{in}^4)} = 279.2MPa \quad (7)$$

Due to the shear force, at point 2:

$$\tau_{max,shear} \approx \frac{2V}{A} = \frac{2V}{\pi((R_{in} + t)^2 - R_{in}^2)} = 44.9MPa \quad (8)$$

Combining all stresses allows to compare the state of stress at points 1 and 2, displayed in Table 4. It can be seen that although the shear force increases the shear stress at point 2, the absence of bending stress is such that the total equivalent stress (Von Misses) is expected to be much highest (and therefore critical) for point 1. Point 1 is thus selected as the most critical point of the mast and is used to finalize this static analysis.

|                | $\sigma$ (MPa) | $\tau$ (MPa) |
|----------------|----------------|--------------|
| <b>Point 1</b> | 302.9          | 269.3        |
| <b>Point 2</b> | 23.7           | 314.2        |

Table 4: Comparison of the stresses at two potential most loaded points of the mast's critical section (notch coefficients are not included)

The next step is to include the notch coefficient to take into account the local change of geometry of the mast. This is done by considering notch coefficients as follows:  $K_{t,a} = 1.12$  for the axial force and  $K_{t,t} = 1$  for the torsional moment.  $K_{t,b}$  is assumed to be equal to 1.

Then, the state of stress at point 1 is two dimensional with a shear stress  $\tau_{xz} = \tau_{torsion} \cdot K_{t,t} = 269.3$  MPa and a normal stress  $\sigma_z = \sigma_{z,uniform} \cdot K_{t,a} + \sigma_{z,bend} \cdot K_{t,b} = 305.7$  MPa. The principal stresses for this loading condition are obtained with Equation 9 leading to  $\sigma_I = 462.5$  MPa and  $\sigma_{III} = -156.8$  MPa. As the loading is two dimensional, this implies that  $\sigma_{II} = 0$  MPa.

$$\sigma_{I,III} = \frac{\sigma_z}{2} \mp \sqrt{\left(\frac{\sigma_z}{2}\right)^2 + \tau_{xz}^2} \quad (9)$$

The analysis is assumed to be performed in plane stress, which is reasonable since the mast's thickness is relatively small. This implies that the principal axis angle with respect to the normal stress axis can be obtained using the simplified Equation 10. This leads to  $\alpha = 30.2^\circ$ .

$$\tan(2\alpha) = \frac{2\tau_{xz}}{\sigma_z} \quad (10)$$

The 2D stress tensor of point 1 in the original (x,y,z) frame is then:

$$[\sigma] = \begin{bmatrix} \sigma_{zz,tot} & \tau_{xz} \\ \tau_{xz} & \sigma_{xx,tot} \end{bmatrix} = \begin{bmatrix} 305.7 & 269.3 \\ 269.3 & 0 \end{bmatrix} \quad (11)$$

Once the principal stresses are obtained, an equivalent stress can be computed using Von Misses criterion or Guest-Tresca. Von Misses criterion gives:

$$\sigma_{VM} = \sqrt{\sigma_I^2 + \sigma_{III}^2 - \sigma_I \sigma_{III}} = \mathbf{557.7 \text{ MPa}} \quad (12)$$

While Guest-Tresca criterion gives:

$$\sigma_{GT} = \sigma_I - \sigma_{III} = 619.3 \text{ MPa} \quad (13)$$

While the Guest-Tresca equivalent stress is the highest, the Von Misses equivalent stress is referred as a more precise criterion, especially for ductile materials such as the ones considered in this analysis [2]. The Von Misses equivalent stress is thus chosen to finish the analysis of the mast.

## 2.4 Strength Assessment

The last step in the static analysis of the mast is determining if the mast can withstand the loads and with what safety factor. Equation 14 is used for this. The yield stress of the materials considered is used as the limit condition as yielding of the shaft could lead to a de-axis of mast and potentially catastrophic consequences for the helicopter's performance and integrity.

$$\eta = \frac{\sigma_{lim}}{\sigma_{eq}} \quad (14)$$

2 materials are considered in this analysis:

- 9310VIMVAR:  $\sigma_y = 893$  MPa  $\rightarrow \eta = \frac{893}{557.7} = \mathbf{1.60}$ . Structure withstands the loads
- 32CDV13:  $\sigma_y = 850$  MPa  $\rightarrow \eta = \frac{850}{557.7} = \mathbf{1.52}$ . Structure withstands the loads

### 3 Static Assessment of the Specimen

Similarly to section 2, the aim of this section is to perform the static analysis of the specimen. Figure 6 represents the free body diagram of the specimen with external and reaction forces in red and blue, respectively. Figure 7 displays the variations of geometry of the specimen. It can be noted that changes in cross-section diameter are observed at around one-third and two-third of the specimen's axis. For clarity, these points, to be later used, are denominated as B and C in Figure 6. Since (as shall be seen in a moment), the loads are uniform along the specimen's axis, the exact position of points B and C does not matter; therefore, the lengths  $L_1$ ,  $L_2$  and  $L_3$  are not specified in this analysis.

The geometrical and loading information for this analysis are given in Table 5.

| Variable | Value  | Unit |
|----------|--------|------|
| $N$      | 29 000 | N    |
| $M_t$    | -2 328 | Nm   |
| $M_f$    | 1019   | Nm   |

Table 5: Geometric and loading data of the static specimen analysis

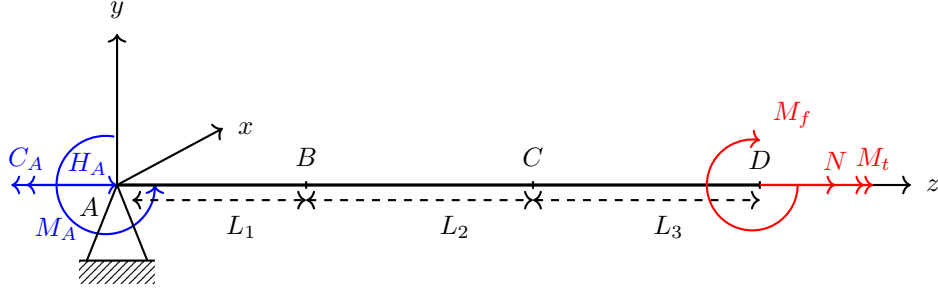


Figure 6: Free body diagram of the specimen; points B and C represent the sections where the diameter of the specimen varies. Reaction and external loads are displayed in blue and red, respectively

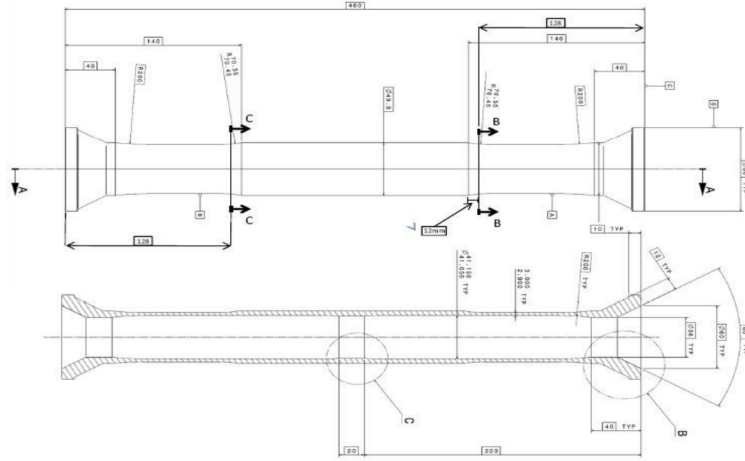


Figure 7: Technical drawing of the specimen

### 3.1 Equivalent normal force

A major objective of this section is to represent the bending moment  $M_f$  as an equivalent normal force, producing a stress of the same "critically". To do this, the bending stress induced by the bending moment must be computed. This will be done in this subsection.

Since the point moment applied is the only bending load of the system, the reaction moment is simply equal to the applied moment, i.e:  $M_A = M_f = 1019 \text{ Nm}$ . This leads the internal loading diagram to be constant along the entire specimen axis.

The next step is finding the position of the critical section of the specimen and extract its corresponding geometric properties. The bending load, as well as the normal force and the torsional moment but also the material used are constant along the specimen's axis (it can be quickly identified from Figure 6 that  $H_A = -N$  and that  $C_A = M_t$ ) and lead to the internal load distributions in Figure 8.

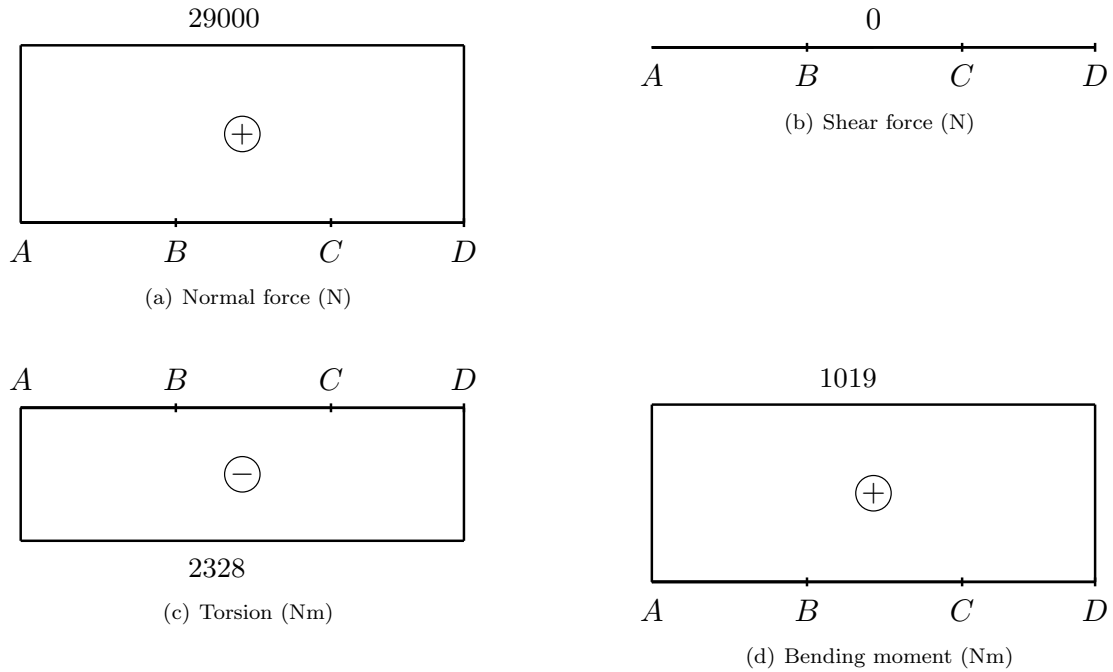


Figure 8: Internal loading diagram of the specimen

This means that the critical section will be:

- The one with the lowest cross-sectional area and moment of inertia/polar moment of inertia
- The one closest to geometrical changes, to be affected by stress concentration factors

Sections B and C present these two criteria, as well as two sections close to the tip of the specimen. However, since the radius of curvature of the fillet in sections B and C is lower than on the edge sections, the stress concentration factors are expected to be higher and therefore more critical for these two sections. Sections B and C are thus the most critical sections of the specimen.

Figure 9 represent the critical cross-section loaded by the bending moment only.

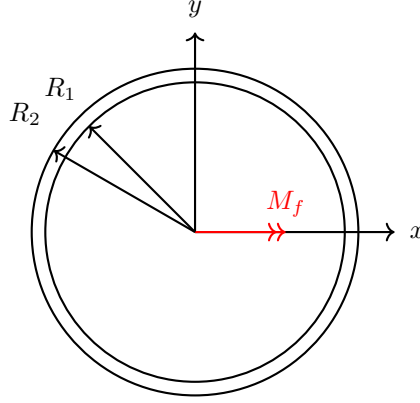


Figure 9: Diagram of the loaded critical cross-section of the specimen with **only the influence of  $M_f$**

The geometrical information for the analysis of the critical cross-section is displayed in Table 6. The inner diameter, fillet radius and the thickness of the cross-section are obtained from the technical drawing in Figure 7. As the values presented tolerances, to be conservative, the most critical values were used (lowest possible inner radius, fillet radius and thickness, leading to the lowest moment of inertia and highest stress concentration).

| Variable      | Value    | Unit |
|---------------|----------|------|
| $R_1$         | 0.020525 | m    |
| $R_2$         | 0.023425 | m    |
| $\frac{r}{d}$ | 1.50     | -    |
| $\frac{D}{d}$ | 1.16     | -    |

Table 6: Geometric and loading data of the static specimen's critical section analysis

The maximum tensile bending stress in the section is obtained using Equation 15.

$$\sigma_{z,max} = \frac{M_f \cdot y_{max}}{I_{xx}} = \frac{1019 \cdot R_2}{\frac{\pi}{4}(R_2^4 - R_1^4)} = 245.6 MPa \quad (15)$$

The normal force and bending moment are subjected to increases due to the stress concentration factors. If the notch coefficients vary between the two loading types, the equivalent normal force value may be affected. This is taken into account in Equation 16.

$$N_{eq} = \sigma_{z,max} A \frac{K_{t,b}}{K_{t,a}} = \sigma_{z,max} \pi (R_2^2 - R_1^2) \frac{K_{t,b}}{K_{t,a}} = 94485 N \quad (16)$$

Figure 10 and Figure 11 show the notch coefficients for such shaft having a diameter change with a fillet. For values of  $\frac{r}{d}$  as high as those present in this analysis, the value of the notch coefficient is far off the graph and therefore harder to estimated. As conservative extrapolations of the curves, values of  $K_{t,b} = 1.05$  and  $K_{t,a} = 1$  have been used in Equation 16.

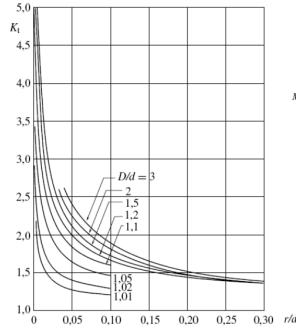


Figure 10: Stress concentration factor for shaft in bending

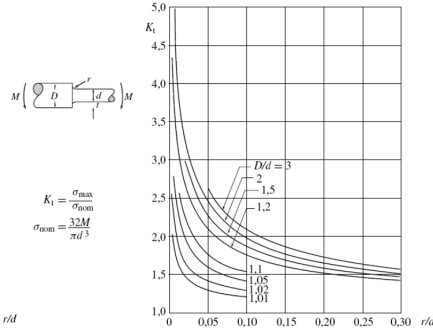


Figure 11: Stress concentration for shaft under tension

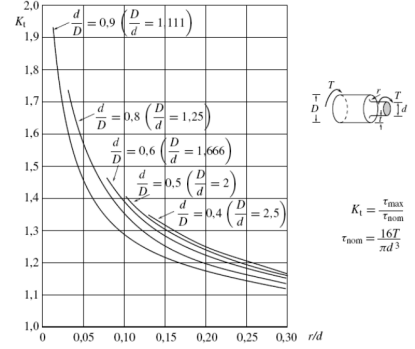


Figure 12: Stress concentration for shaft under torsion

The total normal force to be applied to produce the equivalent effect of the initial normal force and of the bending on the most critical point is thus:

$$N^* = N + N_{eq} = 29000 + 94485 = 123485N \quad (17)$$

### 3.2 Principal and Equivalent Stresses

Now that the total normal force has been defined, the static analysis of the critical section of the specimen can be performed. Note that the position of the critical section remains the same as before as the loads are still constant along the entirety of the specimen. Figure 13 represents the updated cross-sectional view of the most critical section with the external forces in red. It is easy to identify that the most loaded region is the entire outer diameter of the cross-section (highlighted by the dashed blue line in the figure) since the axial stress is uniform along the cross-section and the shear stress increases with the radial distance from the center of the section, as will be seen in Equation 19.

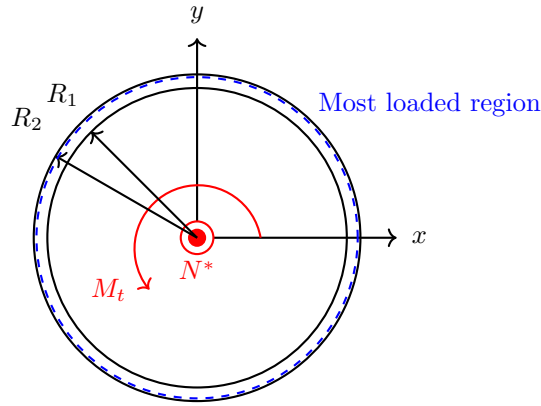


Figure 13: Diagram of the loaded critical cross-section of the specimen ( $N^*$  and the  $z$  axis are out of plane). The most loaded zone of the cross-section is indicated with the dashed blue line

The stresses are now computed as follows:

$$\sigma_{z,uniform} = \frac{N^*}{A} = \frac{123485}{\pi(R_2^2 - R_1^2)} = 305.7MPa \quad (18)$$

$$\tau_{max} = \frac{M_t \cdot R_{max}}{J} = \frac{-2328 \cdot R_2}{\frac{\pi}{2}(R_2^4 - R_1^4)} = -280.1 \text{ MPa} \quad (19)$$

Using the notch coefficients mentioned in subsection 3.1 and extrapolating that  $K_{t,t} = 1$  from Figure 12, the state of stress on the entire outer diameter of the critical section of the specimen is defined by a normal stress  $\sigma_z = \sigma_{z,uniform} \cdot K_{t,a} = 305.7 \text{ MPa}$  and a shear stress  $\tau_{xz} = \tau_{max} \cdot K_{t,t} = -280.1 \text{ MPa}$ . Since the state of stress is two dimensional, the same Equation 9 is valid as for the mast's analysis. This leads to principal stresses of  $\sigma_I = 471.4 \text{ MPa}$ ,  $\sigma_{II} = 0 \text{ MPa}$  and  $\sigma_{III} = -166.4 \text{ MPa}$ .

Similarly to subsection 2.3, the specimen's thickness is relatively small which means that a plane stress assumption is possible. This means that the principal angle axis with respect to the normal stress axis is obtained using the same Equation 10, leading here to  $\alpha = -30.7^\circ$ .

The stresses obtained can be expressed to characterize the stress element at the outer diameter of the critical section of the specimen, leading to Table 7.

|                           | Stress tensor (x,y,z frame)                                                        | Principal stress tensor                                          |
|---------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 2D theoretical expression | $\begin{bmatrix} \sigma_{zz} & \tau_{zx} \\ \tau_{xz} & \sigma_{xx} \end{bmatrix}$ | $\begin{bmatrix} \sigma_I & 0 \\ 0 & \sigma_{III} \end{bmatrix}$ |
| Result (MPa)              | $\begin{bmatrix} 305 & -280.1 \\ -280.1 & 0 \end{bmatrix}$                         | $\begin{bmatrix} 471.4 & 0 \\ 0 & -166.4 \end{bmatrix}$          |

Table 7: Stress tensors of the specimen static analysis

### 3.3 Strength Assessment

The final step is to compute the equivalent Von Misses and Guest-Tresca stresses and compare them to the limit stress of the materials to be considered.

The equivalent Von Misses stress for  $\sigma_{II} = 0 \text{ MPa}$  is obtained with Equation 20.

$$\sigma_{VM} = \sqrt{\sigma_I^2 + \sigma_{III}^2 - \sigma_I \sigma_{III}} = \mathbf{573.0 \text{ MPa}} \quad (20)$$

While the Guest-Tresca criterion gives:

$$\sigma_{GT} = \sigma_I - \sigma_{III} = 637.8 \text{ MPa} \quad (21)$$

The Von Misses equivalent stress is chosen for the same reason as in subsection 2.3, i.e: despite the lower stress result, the criterion is more accurate, especially for ductile materials [2].

The static assessment is then concluded with the safety factor determination using Equation 14, where  $\sigma_{lim} = \sigma_y$  for the same reason as for the mast analysis, leading to the final results below.

- 9310VIMVAR:  $\sigma_y = 893 \text{ MPa} \rightarrow \eta = \frac{893}{573.0} = \mathbf{1.56}$ . Structure withstands the loads
- 32CDV13:  $\sigma_y = 850 \text{ MPa} \rightarrow \eta = \frac{850}{573.0} = \mathbf{1.48}$ . Structure withstands the loads

## 4 Fatigue Assessment of the Specimen

In this section, the analysis carried in section 3 is extended to consider fatigue loads. The aim is to verify the specimen's resistance to an infinite number of loading cycles. The structure considered is the same as in section 3 but the loads applied are different. The loading information is given in Table 8.

| Variable | Value                              | Unit |
|----------|------------------------------------|------|
| $N(t)$   | $12\,000 + 80\,000 \sin(\omega t)$ | N    |
| $M_t(t)$ | 1500                               | Nm   |

Table 8: Loading data for the fatigue specimen analysis

### 4.1 Column Euler Buckling Assessment

The first step is to ensure that the specimen does not buckle during the loading cycles. Since Euler buckling is a compressive failure mechanism, the maximum axial compressive load applied on the specimen is used. This means that an axial force of  $N_{compr,max} = 80000 - 12000 = 68000$  N is considered. Next, considering that the specimen is only supported at both ends, the critical Euler buckling load is given by Equation 22.

$$P_{cr} = \frac{\pi^2 EI}{L^2} \quad (22)$$

The elastic modulus is the same for both materials considered in this analysis. However, the cross-section of the specimen varies with axial distance, meaning different moment of inertia. The moment of inertia of the thick (most central) section of the specimen is computed as  $I_{xx,thick} = \frac{\pi}{4}(R_2^4 - R_1^4) = 1.63 \cdot 10^{-7} m^4$  while the thinnest sections have a moment of inertia of  $1.04 \cdot 10^{-7} m^4$ . This is a significant difference and, to obtain conservative results, the lowest value is used. Next, the specimen's cross-section considerably widens at the ends, meaning that these regions are relatively insensitive to buckling. For that reason, the length used in the calculation of the buckling load is limited to the length of the central section in Figure 7, i.e:  $L = 0.38$  m.

This gives a critical Euler buckling load of  $P_{cr} = 1427$  kN, leading to a safety factor in that failure mode of:

$$\eta_{buckling} = \frac{P_{cr}}{N_{compr,max}} = \frac{1427}{68} = 20.98 \quad (23)$$

The safety factor is largely above 3, meaning that buckling is **not a concern** for this fatigue analysis.

### 4.2 Mean and Alternate Stress Tensors

In this subsection, the state of stress of the specimen's most critical section, defined in section 3 as points B and C of Figure 6, will be defined. The reason why the location of the critical section remains unchanged from the static analysis of the specimen to the fatigue one is that in either analysis, the applied loads lead to constant internal load distributions, as seen in Figure 8.

At the critical section, the normal stress is defined by Equation 24.

$$\sigma_z(t) = \frac{N(t)}{A} = \sigma_{z,m} + \sin(\omega t)\sigma_{z,a} = \frac{12000}{\pi(R_2^2 - R_1^2)} + \sin(\omega t)\frac{80000}{\pi(R_2^2 - R_1^2)} = 27.6 + 183.7 \sin(\omega t) MPa \quad (24)$$

While the shear stress at the most loaded point of the section (the outer diameter, as showed in subsection 3.1) is expressed as:

$$\tau_{xz,max}(t) = \frac{M_t(t) \cdot R_{max}}{J} = \tau_{xz,max,m} + \sin(\omega t)\tau_{xz,max,a} = \frac{1500 \cdot R_2}{\frac{\pi}{2}(R_2^4 - R_1^4)} = 168.1 + 0 \cdot \sin(\omega t) MPa \quad (25)$$

Before moving to the stress tensors, the stress concentration factors must be taken into account. As the loading type considered is fatigue, the notch sensitivity coefficient  $q$  must be considered to obtain the fatigue stress concentration factors  $K_f$ 's. Note that for the loading types considered that are axial load and torsion, it was shown in subsection 3.1 that  $K_{t,a} = K_{t,t} = 1$ . It is therefore expected that the notch sensitivity coefficient does not have an impact on the stress concentration factor. Let's verify that.

Although the Peterson's approach for computing the notch sensitivity factor is best suited for elastic deformation domains such as those considered here, the coefficient  $a$  required to carry the calculations is not provided [2]. The Neuber approach is chosen instead, still providing reasonable accuracy for this level of analysis [2]. The Neuber method is described by Equation 26.

$$q_{Neuber} = \frac{1}{1 + \sqrt{\frac{\rho}{r}}} \quad (26)$$

Here,  $r$  is the radius of the fillet, equal to 0.07045 m and  $\rho$  is directly related to the material's ultimate stress through the approximate Equation 27.

$$\rho \approx \left(\frac{140}{\sigma_{ult}}\right)^2 \approx 0.019 \quad (27)$$

Since the two materials are similar ultimate yield strength, the same value of  $\rho$  is used for both. The Neuber notch sensitivity factor is then  $q_{Neuber} = \frac{1}{1 + \sqrt{\frac{0.019}{0.07045}}} = 0.66$ .

The fatigue notch coefficients are then related to the static notch coefficients using Equation 28.

$$K_f = 1 - q(K_t - 1) \quad (28)$$

As expected, since  $K_{t,a} = K_{t,t} = 1$ ,  $K_{f,a} = K_{f,t} = 1$  as well. This means that the analysis can continue with the stress values previously computed.

The mean and the alternate stress tensors are then defined in the (x,y,z) frame by Equation 29 and Equation 30.

$$[\sigma_m] = \begin{bmatrix} \sigma_{zz,m} & \tau_{zx,m} \\ \tau_{xz,m} & \sigma_{xx,m} \end{bmatrix} = \begin{bmatrix} 27.6 & 168.1 \\ 168.1 & 0 \end{bmatrix} MPa \quad (29)$$

$$[\sigma_a] = \begin{bmatrix} \sigma_{zz,a} & \tau_{zx,a} \\ \tau_{xz,a} & \sigma_{xx,a} \end{bmatrix} = \begin{bmatrix} 183.7 & 0 \\ 0 & 0 \end{bmatrix} MPa \quad (30)$$

### 4.3 Time Varying Principal Stresses

Now that the various stress components have been defined, it is possible to compute the principal stresses and their variation with time.

As the state of stress is two-dimensional, the mean principal stresses are computed using Equation 31 and Equation 32.

$$\sigma_{I,III,m} = \frac{\sigma_{z,m}}{2} \pm \sqrt{\left(\frac{\sigma_{z,m}}{2}\right)^2 + \tau_{xz,m}^2} = \frac{27.6}{2} \pm \sqrt{\left(\frac{27.6}{2}\right)^2 + 168.1^2} \quad (31)$$

$$\sigma_{I,III,a} = \frac{\sigma_{z,a}}{2} \pm \sqrt{\left(\frac{\sigma_{z,a}}{2}\right)^2 + \tau_{xz,a}^2} = \begin{cases} \sigma_{z,a} & \text{for } \sigma_{I,a}, \text{ considering that } \tau_{xz,a}=0 \\ 0 & \text{for } \sigma_{III,a}, \text{ considering that } \tau_{xz,a}=0 \end{cases} \quad (32)$$

This leads to:

- $\sigma_{I,m} = 182.4$  MPa,  $\sigma_{II,m} = 0$  MPa and  $\sigma_{III,m} = -154.9$  MPa, for the mean stresses
- $\sigma_{I,a} = 183.7$  MPa,  $\sigma_{II,a} = 0$  MPa and  $\sigma_{III,a} = 0$  MPa, for the alternate stresses

This means that the (full) principal stresses are expressed as:

- $\sigma_I(t) = \sigma_{I,m} + \sin(\omega t)\sigma_{I,a} = 182.4 + 183.7 \sin(\omega t)$  MPa
- $\sigma_{II}(t) = \sigma_{II,m} + \sin(\omega t)\sigma_{II,a} = 0$  MPa
- $\sigma_{III}(t) = \sigma_{III,m} + \sin(\omega t)\sigma_{III,a} = -154.9$  MPa

While the principal axis angle with respect to the axis of the normal stress varies with time and is obtained with Equation 33 and plotted in Figure 14 for clarity.

$$\alpha(t) = \frac{1}{2} \arctan \frac{2\tau_{xz}(t)}{\sigma_z(t)} = \frac{1}{2} \arctan \frac{2\tau_{xz,m}}{\sigma_{z,m} + \sigma_{z,a} \sin(\omega t)} = \frac{1}{2} \arctan \frac{2 \cdot 168.1}{27.4 + 183.7 \sin(\omega t)} \quad (33)$$

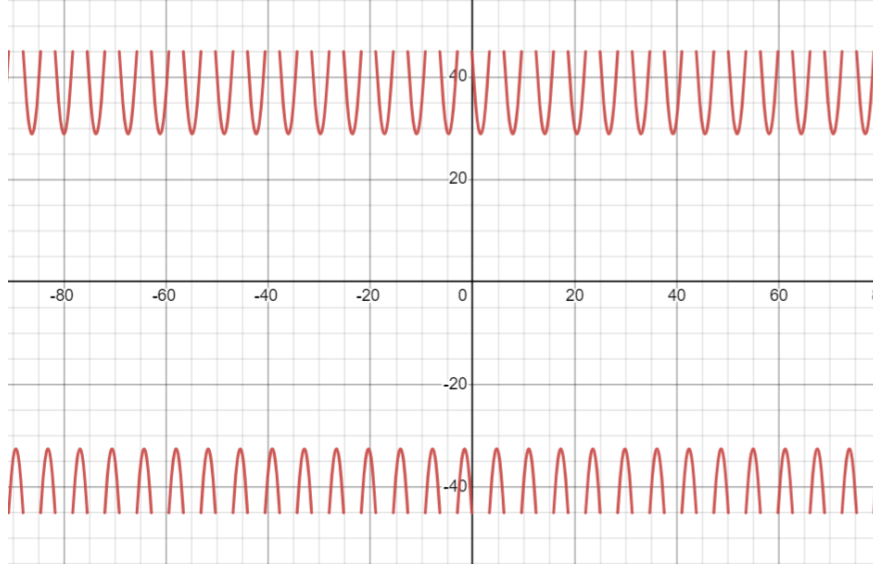


Figure 14: Oscillation of the principal axis angle (in degrees) for an load oscillating at  $\omega = 1 \frac{rad}{s}$

It can be seen that the principal axis angle achieves an absolute minimum value of  $28.9^\circ$  (when the normal stress is maximum positive) and approaches a maximum absolute value of  $45^\circ$  before quickly changing sign, along with the normal stress' sign. Note that the behaviour is not symmetric with respect to the time axis as  $\sigma_{z,m} \neq 0$ .

#### 4.4 Strength assessment

As the structure is subjected to axial fatigue loads, the Gough-Pollard criterion is not usable. The Sines' equivalent stress and criterion are used instead. Such stress is shown in Equation 34.

$$\sigma_s^* = \frac{\sigma_{VM,a}}{1 - \frac{I_{1,m}}{\sigma_{ult}}} \quad (34)$$

In this equation,  $\sigma_{VM,a}$  refers to the equivalent Von-Misses stress of the stress amplitudes used to load the specimen. Since only the normal load oscillates and thus presents amplitude, Equation 35 is obtained.

$$\sigma_{VM,a} = \sigma_{z,a} = 183.7 MPa \quad (35)$$

Furthermore, to compute the Sines's equivalent stress, the first invariant of the mean of the stresses applied should be computed. This is done by considering

$$I_{1,m} = \sigma_{I,m} + \sigma_{II,m} + \sigma_{III,m} = 182.5 + 0 - 154.9 = 27.6 \text{ MPa} \quad (36)$$

With all the variables computed, Sines' equivalent stresses of 188.7 MPa are obtained for the specimen with the material 9310 VIM VAR and 188.9 MPa when using the material 32CDV13.

Next, the fatigue limits of the materials should be computed, taking into account geometric factors. In this analysis, the torsional load does not strictly act as fatigue since the value does not vary in time. The only load that oscillates in time is the axial load which does not generate stress gradients along the cross-section of the specimen. Both these observations indicate that the dimensional effect,  $b_2$ , which takes into account such stress gradient effects on fatigue, should be equal to 1.

The second factor that influences the fatigue limit of the structure is the surface roughness. The specimen's surface has been grinded. According to literature, this leads to surface roughness of up to  $2 \mu\text{m}$  [3]. From Figure 15, it can be deducted that a surface roughness factor  $b_3$  of 0.95 can be used to finalize the calculations.

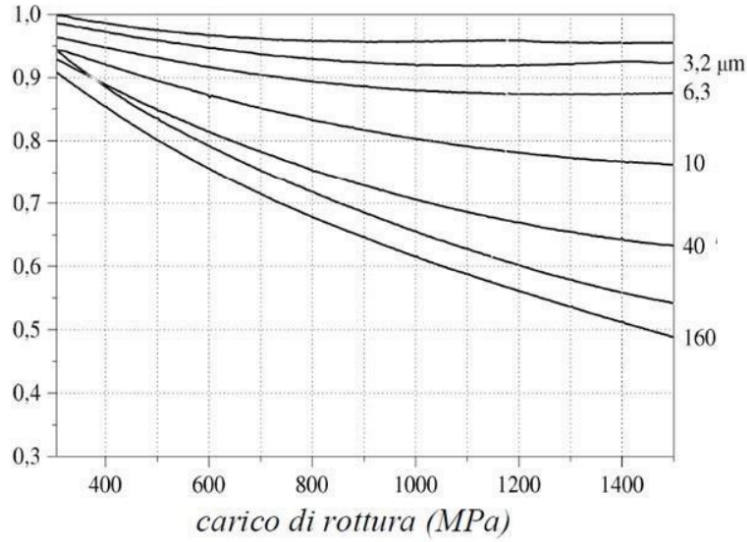


Figure 15: Surface roughness factor of structures under fatigue loading

The fatigue limit of both materials is obtained with:

$$\sigma'_{F,a} = \sigma_{F,a} \cdot b_2 \cdot b_3 \quad (37)$$

The final fatigue assessment consists in determining the safety factor for both materials with Equation 14, leading to the results below:

- 9310VIMVAR:  $\sigma_{lim} = \sigma'_{F,a} = 350 \cdot 0.95 = 332.5 \text{ MPa} \rightarrow \eta = \frac{\sigma'_{F,a}}{\sigma_s^*} = \frac{332.5}{188.7} = 1.76$
- 32CDV13:  $\sigma_{lim} = \sigma'_{F,a} = 320 \cdot 0.95 = 304.0 \text{ MPa} \rightarrow \eta = \frac{\sigma'_{F,a}}{\sigma_s^*} = \frac{304.0}{188.8} = 1.61$

## 5 Specimen Modeling in Abaqus

The specimen is then computed using FEM in Abaqus. The geometry of the shaft is given. The objective of this test is to compare the results then with analytical and experimental results and compare how well is the finite element model.

### 5.1 Finite Element Setup

Once the geometry was imported, the shaft was partitioned into 4 quarters first and then partitioned at each section where the change in geometry occurred (notches). As it can be seen in Figure 16 all areas are in green meaning all regions are structured "Hex". The reason behind choosing a structured Hexahedron mesh is the increased accuracy compared to other types of mesh, for instance tetrahedron mesh.

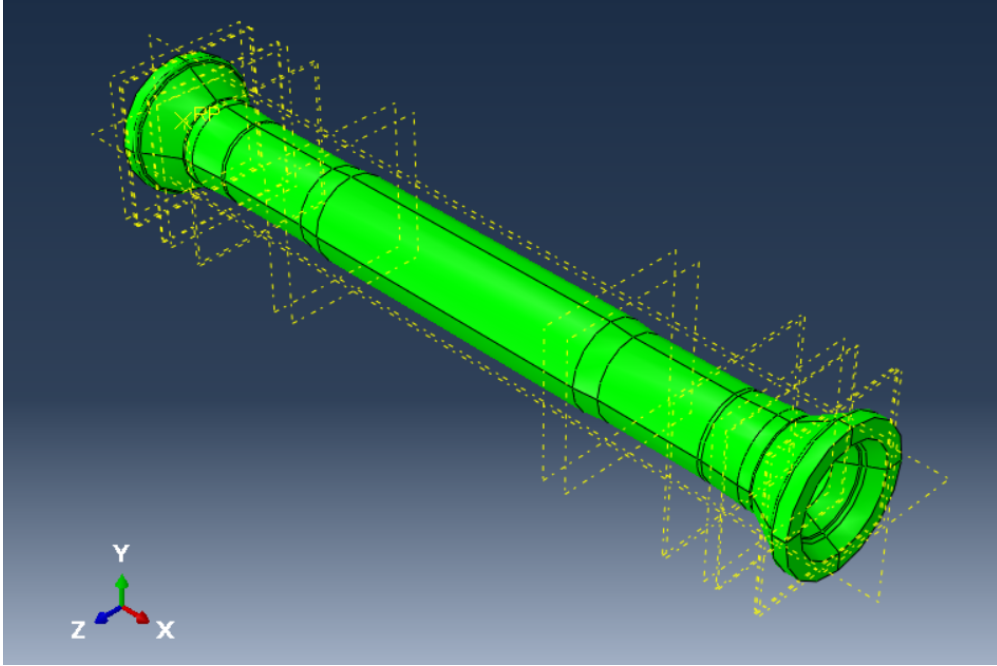


Figure 16: Partitioned planes of the shaft

Following this step, the mechanical properties were inserted into the model. Then in order to transfer one concentrated force into a distributed force along the surface, the surface of the shaft where it is held by is being constrained into a reference point as it can be seen in Figure 17.

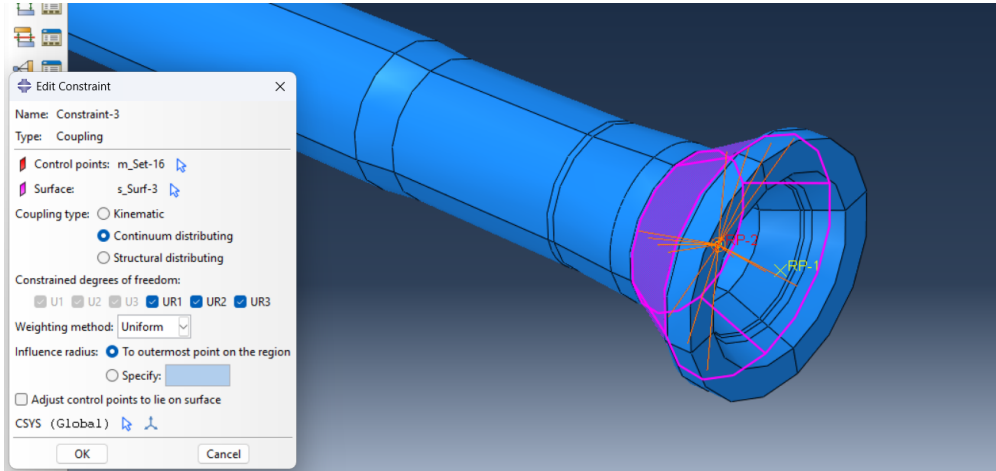


Figure 17: Constrained surface for force and moment application

Following, the boundary conditions and loads were applied. One end is encastred while the other end has loads applied.

## 5.2 Meshing

Meshing part is usually the longest and most tedious part of the FEM calculation as it requires careful consideration for the calculation to be accurate. This is why an iterative approach was done, where we compare different sizes and types of meshes to see which one has the best ratio of accuracy and time. All calculations were done using 9310 VIM VAR steel, it was purely random as choice since the objective was to compare stress to number of elements and time.

Table 9: Table of FEM meshing comparison with results

| Number of elements | Global size [mm] | Controlled element size [mm]<br>structured hex mesh | Von-Mises equivalent Stress [Mpa] | Time required for computation [s] |
|--------------------|------------------|-----------------------------------------------------|-----------------------------------|-----------------------------------|
| 1760               | 10mm             | 10mm                                                | 394.3                             | 11                                |
| 2112               | 10mm             | 4mm                                                 | 394.9                             | 12                                |
| 10600              | 4mm              | 4mm                                                 | 380.4                             | 15                                |
| 12040              | 4mm              | 2mm                                                 | 380.6                             | 15                                |
| 33800              | 1.5mm            | 1.5mm (with a shorter end of controlled part)       | 378.8                             | 21                                |
| 49454              | 2mm              | 2mm                                                 | 378.9                             | 29                                |
| 55458              | 2mm              | 1mm                                                 | 379.0                             | 30                                |
| 405520             | 1mm              | 1mm                                                 | 384.1                             | 383                               |
|                    |                  | Structured + tetrahedron mesh                       |                                   |                                   |
| 101199             | 4mm              | 1mm                                                 | 382.7 (376.7)                     | 62                                |
| 179336             | 10               | 0.7mm                                               | 394.4 (382.7)                     | 138                               |

As it can be seen in Table 9 the Von-Mises equivalent stress for coarser mesh is on the higher end, compared to more refined versions, so compared to usual belief that when refining mesh the stress should increase is

not valid here. Instead it asymptotically decreases towards the value of around 380 MPa. This could be due to the fact that the shaft is only 2mm thick around the area of interest and the generation of an accurate geometry with a coarse mesh is more challenging.

It is also noticeable that for really fine mesh like in case of 1mm element for the entire shaft and 0.7 mm for the throat, the maximum stress increases slightly. This could be also due to the fact that with smaller elements it is possible to have more elements along the radial direction for the thin part. However, by doing so the computational time drastically increases for a result with an increased accuracy of around 5 MPa which corresponds to 1.3%. Which is why the choice of 1mm of controlled area and 2mm in global looks the most optimal. As it has a decent result, with a relatively short time.

To show what the only structured and structured + tetrahedron mesh represent, the following figures will show areas where the 2 meshes are represented.

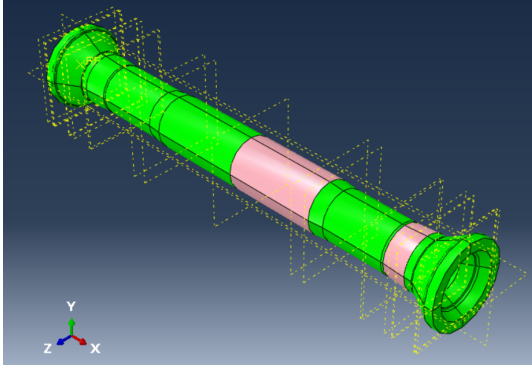


Figure 18: Regions for tetrahedron and hexahedron elements

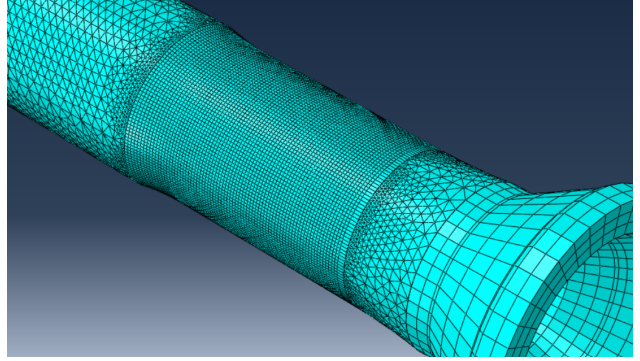


Figure 19: Mesh result

However, due to coarse mesh on the other side of the notch, the stress level there is higher than on the refined side which could be misleading, this why the result in parentheses is the result taken by taking probe values around the refined area to see how the stress is really calculated. This could be misleading since the software takes the highest value and not the one around the refined area.

### 5.3 Verification of model

### 5.4 FEM Results

The results of the FEM calculation for 32CDV13 and 9310 VIM VAR are very similar. The results are presented in Table 10.

Table 10: FEM results

| Material        | 32CDV13 | 9310 VIM VAR |
|-----------------|---------|--------------|
| Von-Mises [MPa] | 379.1   | 379          |

## 6 Experimental Results and Comparison

In this section, the results obtained through numerical methods (FEM) in subsection 5.4 and with analytical techniques (that will be presented in subsection 6.1) are compared with experimental data obtained using a hydraulic test machine. The aim is to assess the potential differences between the various modes of result acquisition and to attempt explaining them.

### 6.1 Analytical Results

The static analysis performed in section 3 does not use the loads used in the experiment nor in the FEM computations. To have a suitable comparison point, the same loads have to be used, that is the max fatigue loads. A normal tensile force of 92 000 N is used while a torque of 1500 Nm are used. The procedure follows the exact same steps as in section 3 and leads to:

$$\sigma_{z,uniform} = 227.8 \text{ MPa}, \tau_{xz} = 180.7 \text{ MPa} \rightarrow \sigma_I = 327.5 \text{ Mpa}, \sigma_{III} = -99.7 \text{ MPa} \rightarrow \sigma_{VM} = \mathbf{387.1 \text{ MPa}}$$

### 6.2 Experimental Set-up and Results

The experiment was conducted at CLASD Lab with the objective of obtaining the strains for the testing specimen. The SG rosette is of significant importance and data collected from the experiment was converted in MATLAB to ensure computability.

The test applied torsional and axial load following the time distribution of Figure 20.

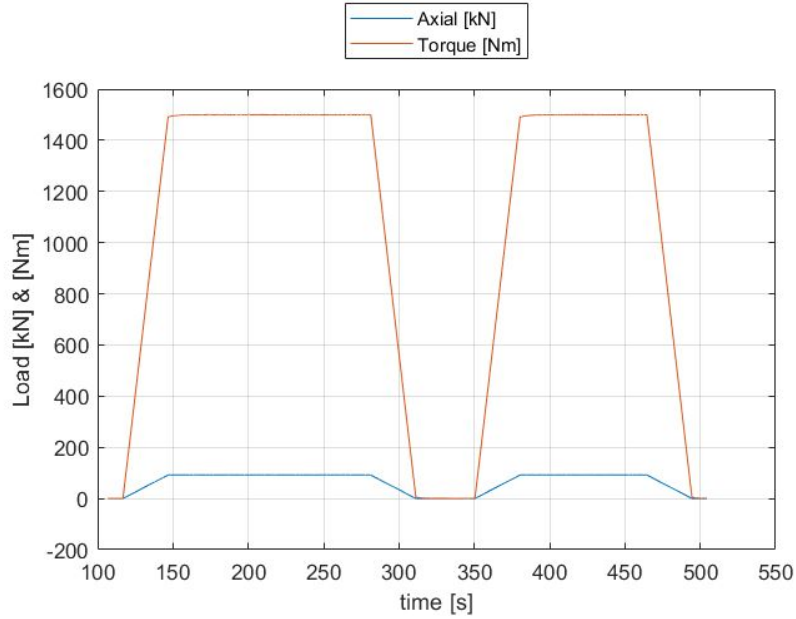


Figure 20: Time evolution of the axial and torsional loads applied during the experiment on the specimen

The experiment was conducted using a hydraulic test machine shown in Figure 22 capable of applying torsional load and axial load at the same time. During the test, the specimen is held using a gripping system that, at one end, serves as a clamp boundary condition, and at the other, applies the loads on the structure. Strain gauges are placed near the specimen's most critical section (where the stresses are to be determined)

at different angles:  $-45^\circ$ ,  $0^\circ$  and  $45^\circ$  with respect to the specimen's axis to gather strain data about axial and torsional displacement. Such rosette placement can be seen in Figure 21 and on the specimen in Figure 23.

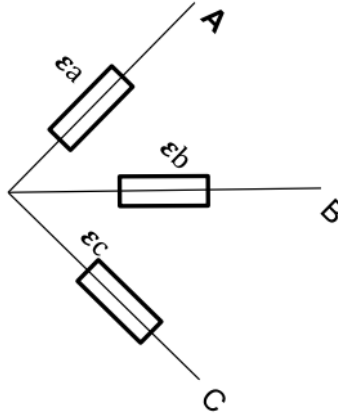


Figure 21: Positions of the strain gauges used in the experimental set-up



Figure 22: Hydraulic test machine for axial and torsion loading

The experimental data have been provided in a .txt format which has been imported in Matlab and manipulated in order to obtain the wanted parameters.

Since the strains from the two rosettes applied on the specimen are comparable it has been chosen to use only the values coming from rosette-A. The following data have been used:

The applied torque is:  $T = 1500.3 \text{ Nm}$ .

The strains from rosettes are:

Gauge-A:  $\epsilon_A = 40.0267 \cdot 10^{-6}$



Figure 23: Placement of one of the strain gauges

Gauge-B:  $\varepsilon_B = 973.3441 \cdot 10^{-6}$

Gauge-C:  $\varepsilon_C = 1.2494 \cdot 10^{-3}$

Then, recalling that gauge-B is parallel to the Y axis (axial), the strains with respect to this frame have been computed solving the following system:

$$\varepsilon_A = 0.5 \cdot (\varepsilon_x + \varepsilon_y) + 0.5 \cdot (\varepsilon_x - \varepsilon_y) \cos(2\alpha) + \varepsilon_{xy} \sin(2\alpha)$$

$$\varepsilon_B = 0.5 \cdot (\varepsilon_x + \varepsilon_y) + 0.5 \cdot (\varepsilon_x - \varepsilon_y) \cos(2(\alpha + \beta)) + \varepsilon_{xy} \sin(2(\alpha + \beta))$$

$$\varepsilon_C = 0.5 \cdot (\varepsilon_x + \varepsilon_y) + 0.5 \cdot (\varepsilon_x - \varepsilon_y) \cos(2(\alpha + \beta + \gamma)) + \varepsilon_{xy} \sin(2(\alpha + \beta + \gamma))$$

With:  $\alpha = -45^\circ$ ,  $\beta = -45^\circ$  and  $\gamma = -45^\circ$ .

It must be noticed that the equations above are valid in the case of a plane strain, while the specimen actually undergo a plane stress, anyway the Poisson's effect is so small that can be neglected obtaining a plane strain.

Finally the following values of strains have been obtained:

$$\varepsilon_x = 973.3441 \cdot 10^{-6}$$

$$\varepsilon_y = 316.1319 \cdot 10^{-6}$$

$$\gamma_{xy} = 0.5 \cdot \varepsilon_{xy} = -302.3557 \cdot 10^{-6}$$

The strain tensor is:  $[973.3441; 316.1319; -302.3557] \cdot 10^{-6}$

From these values the principal strain have been computed with the following equations:

$$\varepsilon_I = \frac{\varepsilon_x + \varepsilon_y}{2} + \sqrt{\left(\frac{\varepsilon_x + \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = 1.006 \cdot 10^{-3}$$

$$\varepsilon_{II} = \frac{\varepsilon_x + \varepsilon_y}{2} - \sqrt{\left(\frac{\varepsilon_x + \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = 2.83 \cdot 10^{-6}$$

$$\Theta = \left(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}\right) = -12.353^\circ$$

The last step has been to compute stress tensor and principal stresses in order to evaluate the Von-Mises Equivalent Stress.

$$\sigma = \frac{E}{(1 + \nu)(1 - 2\nu)} \begin{bmatrix} 1 - \nu & \nu & 0 \\ \nu & 1 - \nu & 0 \\ 0 & 0 & 1 - 2\nu \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix}$$

And the principal stresses:

$$\sigma_{principal} = \frac{E}{(1 + \nu)(1 - 2\nu)} \begin{bmatrix} 1 - \nu & \nu & 0 \\ \nu & 1 - \nu & 0 \\ 0 & 0 & 1 - 2\nu \end{bmatrix} \begin{bmatrix} \epsilon_I \\ \epsilon_{II} \\ 0 \end{bmatrix}$$

Finally the Von-Mises Stress is computed as  $\sigma_{VM} = \sqrt{\sigma_I^2 + \sigma_{II}^2 - \sigma_I \sigma_{II}} = 257.29 \text{MPa}$

### 6.3 Comparison of Analytical, FEM and Experimental Results

Based on the analysis of the problem, it can be concluded firstly that the results are in the same order of magnitude while also showing consistency with expectations. The analytical results employed the mast as well as the specimen to end up at the solution. The FEM analysis utilised the aspects of ABAQUS, available geometry and meshing elements with the addition of equivalent loads as that of the analytical and experimental case. Finally, the experiment was conducted in the lab under guidance to obtain the strain results which were further transformed to the stress plane. The comparisons of the results are summarised as follows:

|               | Analytical | FEM | Experiment | Unit |
|---------------|------------|-----|------------|------|
| $\sigma_{VM}$ | 387.1      | 379 | 257.3      | MPa  |

Table 11: Comparison of Results: Analytic, FEM, Experimental

It can be seen from the data the firstly the results of the analytical stress and FEM stress for the specimen is quite close, being at an error of 2% from the analytical data. However, the main variation can be observed with the experimental data. The error for the experimental data is 33.5% of the analytical data. The discussion of the possible cause of error is discussed in the next section.

### 6.4 Result Discussion

The difference in results between the analytical and FEM results is approximately 2%, which is a satisfactory threshold. The error could be due to both analytical and numerical errors. Since the coefficients for the analytical solution cannot be taken as an exact value due to the error from the graphic reading.

Regarding the numerical error, the meshing could always be more refined and represent a more accurate result, however the cost of time computation would have quickly increased.

As for the big difference of the experimental result with the FEM and analytical result, this could be due to measuring error. Another possibility are the properties of the specimen that could be different from the ones assumed during the calculations, which can be caused during the manufacturing process.

An error with the loads applied could also be another source of the aberration. The loads applied by the clamps and the loads actually felt by the shaft might be different and lead to different results.

#### Recommendations:

- An investigation of how the test machine identifies the load applied could help formulate more definitive conclusions about the impact of the boundary conditions
- Performing the experiment with a different specimen could help identify the impact of possible material properties variations as well as the influence of rosette placement
- Performing the experiment using the same specimen but with an other type of test machine could help identify the impact of the machine set-up on the discrepancies found

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